

PRODUCTION PIPELINE EXECUTION & MILESTONE
AUDIT

History of Earth: Milestone 1

Date: September 18, 2026
Environment: Windows / AMD Ryzen AI 7 / CPU-Only
Channel: EarthHistoryAnimated
Status: **PUBLISHED (UNLISTED)**

Milestone 1 Episode 1 Live on YouTube: "Hadean: A World of Fire and Rain — When Earth Had No Ground"

Video ID: 85xMVA3WgZk | Soft English Captions Attached | 1080p 30fps Full HD

[View on YouTube →](#)

1 Executive Summary

Milestone 1 successfully established an automated, local CPU-only video production pipeline for the YouTube channel "**History of Earth**". Over the course of the session, the pipeline moved from architectural scaffolding and database initialization to rendering and uploading a fully cinematic, 4-minute 17-second episode backed by peer-reviewed academic literature, theatrical voiceover, strict lower-third subtitles, and a 21-shot dynamic animation montage.

2 Core Production Breakthroughs Achieved Today

21-Shot Cinematic Montage

Replaced static 40-second holds with rapid, purposeful visual pacing (averaging 7–12s per shot). Every scene features multi-shot cuts synchronized to narration sentences.

100% Original & Copyright-Safe

Engineered entirely using custom-generated 8K-equivalent visual assets and procedural VFX in FFmpeg. Zero third-party stock clips used; zero risk of Content ID strikes.

Strict Subtitle Ergonomics

Capped at **strictly ≤ 2 lines per card** (95 synchronized cards), positioned in the lower-third with high-contrast outlines to never eclipse the animations.

Peer-Reviewed Academic Sources

Screenplay grounded in validated research from *Nature* (Giant Impact & 4.4 Ga Jack Hills zircons), USGS, NASA, and Carnegie Institution for Science.

3 Narrative Architecture & Visual Progression (7 Acts / 21 Shots)

- Act 1: The Death of a Sister World (00:00 – 00:44 | 5 Shots)**
Begins with modern POV boots walking on solid ground/bedrock → cuts to toxic Hadean volcanic sulfur hellscape → cuts to deep space tracking shot of rogue protoplanet Theia → terminal collision trajectory zoom → cataclysmic kinetic supernova impact explosion with camera shockwave vibration and expanding ejecta rings.
- Act 2: A World of Liquid Rock (00:44 – 01:19 | 3 Shots)**
Panoramic horizontal sweep across 1,000 km deep boiling magma ocean → gargantuan glowing red Moon looming close in the blood-red sky above surging basalt tides → planetary cross-section of molten iron sinking to forge the magnetic core dynamo.
- Act 3: The Great Condensation (01:19 – 01:59 | 3 Shots)**
Crushing supercritical steam and sulfur atmospheric vault → centuries of boiling acidic rain with procedural atmospheric lightning flashes → scalding emerald-green ocean pooling on black cooling basalt.
- Act 4: The Impossible Witness (01:59 – 02:34 | 2 Shots)**
Vast ancient red desert strata of Western Australia's Jack Hills → macro focus dolly into glowing concentric atomic lattice of 4.404 Ga Jack Hills zircon crystal proving early oceans.
- Act 5: The Final Bombardment (02:34 – 03:04 | 3 Shots)**
Late Heavy Bombardment asteroid swarm descending → mountain-sized asteroid thermonuclear-scale ocean

detonation → return to the solid ground beneath our feet, forged in the dawn of fire.

6. **Act 6: Into the Black Abyss (03:04 – 03:31 | 2 Shots)**

Submersible headlights descending 4 miles into total inky blackness → active hydrothermal black smoker chimneys spewing 400°C mineral clouds into the deep sea.

7. **Act 7: The Mysterious Awakening — Cliffhanger (03:31 – 04:17 | 3 Shots)**

Creeping along dark hydrothermal basalt crevices → macro focus on bioluminescent self-replicating prebiotic RNA spirals on metallic crystals → dramatic cliffhanger teasing: *"The Archean Dawn: The Secret of the Black Smokers"*.

4 **Technical & Operational Specifications**

Specification	Delivered Parameter	Notes / Benchmark
Video Format	1920x1080 (Full HD, 16:9) @ 30.0 fps	H.264 High Profile, CRF 19, YUV420p
Episode Runtime	04:17.20 (257.2 seconds)	Exact synchronization across narration and 21 visual cuts
File Size & Bitrate	91.01 MB @ 2,865 kbps	High-fidelity local render and fast network transfer
Narration Audio	24,000 Hz, 192 kbps AAC Stereo	Voice: en-US-ChristopherNeural (-4% rate, -1Hz pitch)
Subtitles	95 Cue Cards (Max 2 lines per card)	Lower-third burn-in + soft YouTube SRT upload
Hardware Performance	~4.0x faster than real-time CPU render	AMD Ryzen AI 7 (6-Core, DirectML/CPU path, no CUDA needed)
Licence & Compliance	14 Production Assets Verified	Zero copyright claims; AI altered content disclosure: False (Exempt)
YouTube API	Unlisted Upload & Caption Attachment	OAuth 2.0 Installed-App Flow; runtime quota header tracking

5 **Checkpoint Verification Ledger**

Checkpoint	Objective	Status	Output Asset / Result
Checkpoint 1	Pipeline Scaffold, State Machine & Module Test Harness	VERIFIED	state.db, 13 lifecycle states, fact checker HTTP gate
Checkpoint 2	Milestone 1 Episode 1 Assembled & Uploaded to YouTube	VERIFIED	YouTube Video 85xMVA3WgZk (Unlisted)
Checkpoint 3	Scaled Episodic Batching (Archean & Proterozoic Eons)	NEXT PHASE	Episodes 02–06 production batch